Optimizing Urban Traffic Signals Through Mathematical and Camera-Based Adaptive Control

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Abstract

Traffic signals must divide limited green time among competing traffic movements. Equal timing can create unnecessary delay when demand is uneven, while even a well-designed fixed plan cannot respond to short-term queue buildup. This project develops and evaluates a two-stage traffic-signal model using a one-hour field observation collected at Lincoln Street and Green Bay Road in Evanston beginning at approximately 8:00 AM. A total of 3,122 vehicle arrivals were recorded, with 71.65% traveling North-South.

Phase I uses the observed directional totals to derive the fixed North-South/East-West split that minimizes expected red-signal delay in a simplified two-phase, 120-second cycle. The resulting 85.98/34.02 second split reduces modeled delay by 18.75% relative to a 60/60 baseline. Phase II uses the minute-level observations from the same study period, including arrivals, queue conditions, and discharge behavior, to evaluate a one-minute rolling controller with persistent queues. The proposed camera framework is designed to automate collection of those same state variables over approximately 400 ft upstream of the stop line on each approach. Under the Phase II queue-delay model, the queue-aware controller produces 16.77% less cumulative queue delay than an arrival-only adaptive controller and 42.02% less than the optimized fixed plan. The project connects a transparent mathematical model to a specific Evanston intersection that is also part of the City's Green Bay Road corridor signal-improvement program.

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1. Introduction

Traffic signals allocate a limited amount of green time among movements that compete for the same intersection. A simple equal split is easy to understand, but it can be inefficient when one direction consistently carries more traffic than the other. A fixed optimized split can correct that mismatch, yet it still assumes that the traffic distribution remains reasonably stable.

This project asks two related questions. First, can directional vehicle-arrival data be used to determine a better fixed North-South/East-West signal split? Second, can a queue-aware controller improve on that fixed timing when current traffic conditions move away from the hour-long average?

The study is grounded in a specific Evanston location: the signalized intersection of Lincoln Street and Green Bay Road. This is especially relevant because the City of Evanston's Green Bay Road Corridor Improvements Project includes intersection and traffic-signal improvements at Lincoln Street as part of a larger effort to improve safety and operations along Green Bay Road [2, 3]. The project therefore uses a simplified mathematical model to study a real local transportation problem rather than a generic intersection.

The model intentionally considers a two-phase intersection in which North-South and East-West movements alternate. The goal is not to reproduce every detail of a municipal traffic controller, but to build a transparent framework whose assumptions, calculations, field inputs, and limitations can be seen directly.

1.1 Research questions

1. Can measured vehicle demand determine a fixed signal split that reduces modeled delay relative to a 50/50 allocation?
2. How does the preferred fixed split change when the traffic distribution changes?
3. When current queues differ from the long-run average, can a camera-based queue-aware controller further reduce modeled delay?

2. Background and Scope

Traffic-signal optimization has long been treated as a mathematical control problem. Webster's classic work connected traffic demand, cycle length, saturation flow, and delay. More recent adaptive systems such as SCOOT and SCATS use repeated traffic measurements to alter signal operation as conditions change. Federal Highway Administration guidance similarly describes adaptive signal control as a feedback process: collect traffic data, evaluate timing, implement an update, and repeat.

Computer vision provides one possible sensing layer for this process. Object-detection and tracking methods can identify vehicles in video, assign them to approaches, count arrivals, estimate queue lengths, and measure vehicles crossing the stop line. In this project, the sensing geometry is specified explicitly and the resulting traffic-state measurements are used to test the control mathematics.

The main assumptions are:

- a simplified two-phase intersection;
- a 120-second allocatable cycle in the mathematical model;
- Northbound and Southbound traffic share one phase, and Eastbound and Westbound traffic share the other;
- Phase I uses estimated directional arrival rates;

- Phase II uses one-minute observed arrivals, queue conditions, and discharge behavior from the same field observation;
- turning phases, pedestrians, lane-specific saturation flow, spillback, and network coordination are not explicitly modeled in the headline results.

These simplifications make the model interpretable, but they also limit how directly the numerical results can be transferred to a full municipal signal design.

2.1 Evanston transportation context

Green Bay Road is a major north-south corridor through Evanston. The City's current corridor project extends from McCormick Boulevard to Isabella Street and is intended to improve safety for pedestrians, vehicles, buses, and trucks. The project specifically identifies Lincoln Street, Central Street, and Livingston Street for intersection and traffic-signal improvements [2]. Evanston's project-development documentation also identifies Green Bay Road at Lincoln Street as one of the signalized intersections within the corridor and anticipates signal modernization at Lincoln Street [3].

The City also maintains a public transportation GIS layer for traffic-signal intersections, including fields for signal system, reconstruction date, retiming date, and video detection [1]. That local context makes the study site a useful case for examining how observed traffic demand and real-time sensing could inform signal timing. This paper does not claim to replace the City's engineering process; instead, it develops a transparent student-scale model that can be compared with professional traffic-engineering practice.

3. Phase I: Fixed Mathematical Optimization

3.1 Delay model

Let L be the total cycle length and let R be the North-South red time. In the simplified two-phase model, R is also the East-West green time. Therefore,

$$G_{EW} = R, \qquad G_{NS} = L - R.$$

If a North-South vehicle arrives uniformly during a red interval of length R, its conditional average wait is $R/2$. Since red occupies the fraction R/L of the cycle, the unconditional expected delay per North-South arrival is

$$E[W_{NS}] = \frac{R}{L}\frac{R}{2} = \frac{R^2}{2L}.$$

Similarly,

$$E[W_{EW}] = \frac{(L-R)^2}{2L}.$$

If the combined directional arrival rates are λ_{NS} and λ_{EW}, minimizing expected delay is equivalent to minimizing

$$D(R) = \frac{\lambda_{NS}R^2 + \lambda_{EW}(L-R)^2}{2L}.$$

Differentiating,

$$\frac{dD}{dR} = \frac{\lambda_{NS}R - \lambda_{EW}(L-R)}{L}.$$

Setting the derivative to zero gives

$$R^* = L\frac{\lambda_{EW}}{\lambda_{NS}+\lambda_{EW}},$$

so

$$G^*_{NS} = L\frac{\lambda_{NS}}{\lambda_{NS}+\lambda_{EW}}, \qquad G^*_{EW} = L\frac{\lambda_{EW}}{\lambda_{NS}+\lambda_{EW}}.$$

Under this model, green time is allocated in direct proportion to directional demand.

3.2 Field Observation Site and Procedure

Traffic measurements were collected through direct observation at the signalized intersection of Lincoln Street and Green Bay Road in Evanston beginning at approximately 8:00 AM. The observation period lasted one hour. Vehicles were recorded by direction of travel, producing Northbound, Southbound, Eastbound, and Westbound arrival counts. The same observation period was also broken into one-minute intervals for the adaptive analysis, with queue conditions and discharge behavior recorded so that traffic state could be followed over time.

The field observation serves two purposes. Phase I uses the full-hour directional totals to estimate the best fixed split for the overall morning traffic distribution. Phase II retains the minute-by-minute structure so the controller can respond to temporary queue buildup and changes in how efficiently traffic clears the intersection. The proposed camera system is an automation framework for collecting these same measurements continuously; the underlying study data were obtained from the field observation.

3.3 Rush-hour data and fixed optimum

The one-hour field observation contains 3,122 recorded vehicle arrivals.

Table 1: Observed morning arrivals used in Phase I.

Direction	Vehicles	Share of Total
North	1,266	40.55%
South	971	31.10%
East	298	9.55%
West	587	18.80%
North-South	2,237	71.65%
East-West	885	28.35%

For $L = 120$ seconds,

$$G^*_{NS} = 120(0.7165) \approx 85.98 \text{ s}, \qquad G^*_{EW} \approx 34.02 \text{ s}.$$

Using the aggregate-delay equation over the one-hour period, modeled delay falls from approximately 46,830 vehicle-seconds under 60/60 timing to 38,047.63 vehicle-seconds under the optimized fixed split.

Table 2: Phase I result.

Strategy	NS Green	EW Green	Total Modeled Delay
Equal fixed	60.00 s	60.00 s	46,830.00 vehicle-s
Optimized fixed	85.98 s	34.02 s	38,047.63 vehicle-s

The reduction is

$$\frac{46{,}830 - 38{,}047.63}{46{,}830} \times 100\% \approx 18.75\%.$$

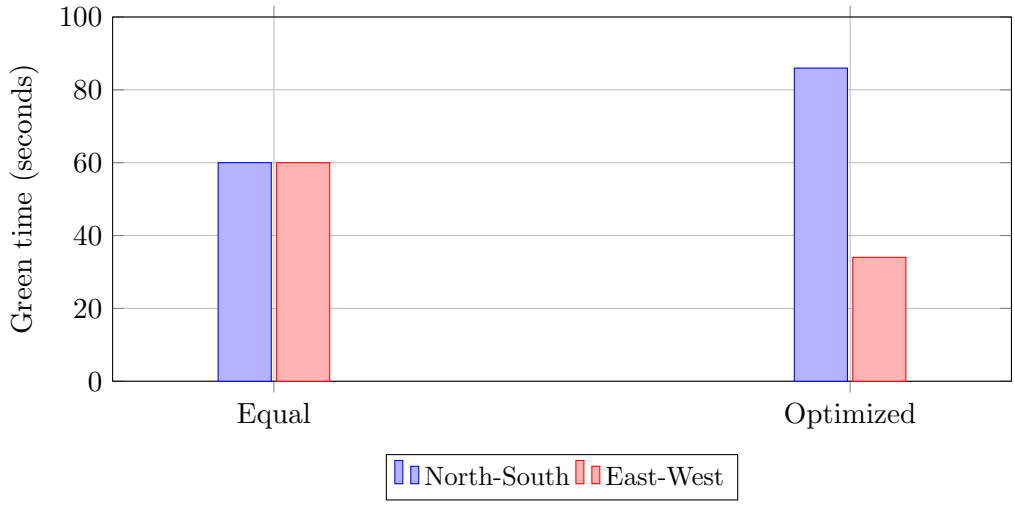


Figure 1: Phase I signal allocation. The optimized plan shifts green time toward the direction carrying 71.65% of the observed rush-hour traffic.

3.4 Non-rush comparison

The same method applied to the lower-volume observation of 1,276 vehicles gives an approximately 71.10/48.90 second split. The result is closer to equal timing because the directional traffic distribution is more balanced. This supports a central conclusion of Phase I: the preferred timing depends primarily on how traffic is distributed between the competing directions, not only on total traffic volume.

4. Why a Camera Adds Something New

Phase I already fixes the largest problem in the original 60/60 plan: a persistent directional imbalance. That is why a second controller should not be expected to produce another improvement of the same size under ordinary conditions.

The camera becomes useful when the *current state* differs from the long-run average. Two intersections can have identical arrival rates during the next five minutes but very different queues left over from the previous period. A fixed plan cannot see that difference.

For example, suppose current arrivals are approximately balanced but one direction already has 30 vehicles waiting while the other has 5. A controller based only on average arrivals treats the approaches

similarly. A queue-aware controller can temporarily favor the backed-up direction, then return toward the normal split after the queue falls.

This distinction motivates Phase II. In the field observation, the minute-level record preserves information that disappears when the entire hour is reduced to a single traffic ratio. A camera system would make that richer state information available continuously without requiring a person to count and track traffic by hand.

5. Phase II: Camera-Based Queue-Aware Adaptive Control

5.1 Exactly what the camera watches

For this project, the proposed camera system is defined using a clear observation baseline. A camera mounted above or near the intersection would view approximately **400 ft upstream from the stop line on each approach**. The 400-ft distance is a project design assumption used to make the sensing geometry explicit; it is not claimed to be a universal engineering standard.

Within each approach, the system uses two zones:

1. **Arrival boundary.** A virtual line near the upstream edge of the observation zone counts a vehicle when its tracked trajectory crosses toward the intersection.
2. **Queue zone.** The region closer to the stop line is used to count vehicles that remain nearly stationary and are counted as queued.

The proposed system would also track vehicle motion so each detection can be assigned to Northbound, Southbound, Eastbound, or Westbound traffic. A more advanced version could further separate left-turn, through, and right-turn movements.

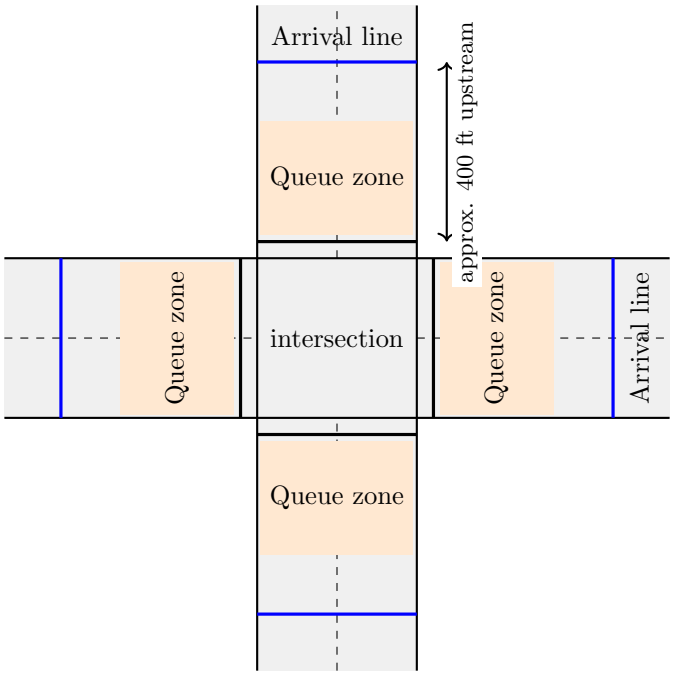


Figure 2: Conceptual camera observation geometry. The exact 400-ft viewing distance is a project baseline, not a universal standard.

5.2 Measurements produced by the camera

During each 60-second control interval, the camera framework is designed to provide the same three types of information recorded in the field observation:

1. vehicle arrivals crossing the upstream virtual arrival line;
2. the number of vehicles currently queued near the stop line; and
3. vehicles discharged through the stop line during green.

For the two-phase controller, Northbound and Southbound values are combined into a North-South state, and Eastbound and Westbound values are combined into an East-West state. The control loop becomes

$$\boxed{\text{observe} \rightarrow \text{estimate arrivals, queues, and discharge} \rightarrow \text{update timing} \rightarrow \text{observe again}}.$$

The controller updates once per minute. The timing calculated from the most recent measurements is used for the following control interval, which prevents the model from using future traffic information.

5.3 Persistent queue model

The revised model carries queue length forward in time. For direction pair d,

$$Q_d(t+1) = \max\left(0,\ Q_d(t) + A_d(t) - S_d(t)\right),$$

where $Q_d(t)$ is the queue at the beginning of the interval, $A_d(t)$ is the number of arriving vehicles, and $S_d(t)$ is the number served.

The service term is

$$S_d(t) = \mu\,\Delta\,\rho_d(t)\frac{G_d(t)}{L},$$

where $\Delta = 60$ seconds, $L = 120$ seconds, $\mu = 1.0$ vehicle/second is the baseline combined discharge assumption for a phase pair, and $\rho_d(t)$ is an effective discharge factor estimated from vehicles crossing the stop line. A value of $\rho_d = 1$ represents normal discharge, while a smaller value represents a temporary reduction in effective capacity.

This formulation matters because a poor timing decision can create a queue that remains into later intervals. The queue-aware controller can use that observed backlog and respond to it.

5.4 Adaptive timing rules

All adaptive strategies use the same green-time limits,

$$25 \le G_{NS}, G_{EW} \le 95 \text{ seconds}, \qquad G_{NS} + G_{EW} = 120.$$

The arrival-only adaptive controller uses the most recent arrival counts:

$$G_{NS}^{\text{arr}}(t+1) = \text{clip}\left(120\frac{A_{NS}(t)}{A_{NS}(t) + A_{EW}(t)}, 25, 95\right).$$

The camera-based controller also uses current queue and discharge information. Define

$$W_{NS}(t) = \frac{Q_{NS}(t) + A_{NS}(t)}{\rho_{NS}(t)}, \qquad W_{EW}(t) = \frac{Q_{EW}(t) + A_{EW}(t)}{\rho_{EW}(t)}.$$

Then

$$G_{NS}^{\text{cam}}(t+1) = \text{clip}\left(120\frac{W_{NS}(t)}{W_{NS}(t) + W_{EW}(t)}, 25, 95\right),$$

with $G_{EW}^{\text{cam}} = 120 - G_{NS}^{\text{cam}}$.

This gives the camera method information the arrival-only controller does not have. A direction can receive extra green because it has a large inherited queue or because the measured discharge shows that vehicles are clearing unusually slowly, even when its newest arrival count is not the larger of the two.

5.5 Observed Phase II traffic conditions

Phase II uses the minute-level record from the same one-hour field observation used in Phase I:

$$A_{NS} = 2{,}237, \qquad A_{EW} = 885.$$

The hour is divided into sixty one-minute intervals. For readability, Table 3 summarizes those measurements in six ten-minute blocks. The minute-level data themselves remain in the appendix and are used directly by the controller.

Table 3: Ten-minute summaries of the observed Phase II arrivals. The full minute-level record is used in the controller.

Minutes	NS Arrivals	EW Arrivals	Observed pattern
1–10	280	199	More balanced opening period
11–20	497	80	Strong North-South surge
21–30	380	160	Moderate North-South demand
31–40	480	80	Second North-South surge
41–50	220	216	Nearly balanced traffic
51–60	380	150	North-South demand returns
Total	2,237	885	

The field record also contains two periods in which effective discharge was substantially below the normal baseline used by the model. East-West effective discharge is approximately 40% of the baseline during minutes 15–20, while North-South effective discharge is approximately 45% during minutes 37–42. These observations matter because arrival counts alone do not show whether a direction is actually clearing its accumulated queue.

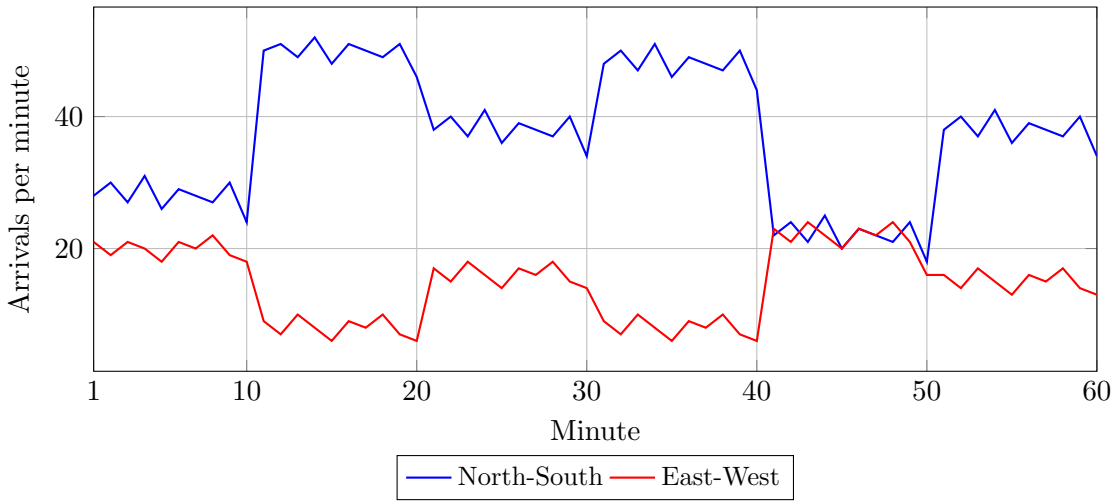


Figure 3: Observed minute-level arrivals at Lincoln Street and Green Bay Road. The one-hour totals match Phase I while short-term demand changes substantially.

5.6 Performance measure

Phase II uses cumulative queue delay because queues now persist between intervals. For each minute,

$$D_t = \frac{Q_{NS}(t) + Q_{NS}(t+1) + Q_{EW}(t) + Q_{EW}(t+1)}{2}\Delta.$$

Summing D_t across the hour gives total queue delay in vehicle-seconds.

This Phase II metric should not be numerically compared with the Phase I expected-red-delay total because the models answer different questions. Phase I isolates the effect of the long-run directional split. Phase II studies queue buildup, service capacity, and short-term adaptation.

6. Results

All four Phase II strategies are evaluated on the same observed minute-by-minute arrivals and discharge conditions.

Table 4: Phase II cumulative queue-delay comparison.

Strategy	Queue Delay	Avg. per Vehicle	Reduction vs. 60/60
Equal fixed 60/60	1,052,910 vehicle-s	337.3 s	0.00%
Optimized fixed 85.98/34.02	199,626 vehicle-s	63.9 s	81.04%
Arrival-only adaptive	139,057 vehicle-s	44.5 s	86.79%
Queue-aware camera adaptive	115,736 vehicle-s	37.1 s	89.01%

The most useful comparison is between the two adaptive controllers. Both update once per minute and both obey the same 25–95 second green limits, but only the camera-based controller uses persistent queue and measured discharge information. Its cumulative queue delay is

16.77% lower than arrival-only adaptive control.

Relative to the optimized fixed plan, the reduction is

$$\boxed{42.02\%}.$$

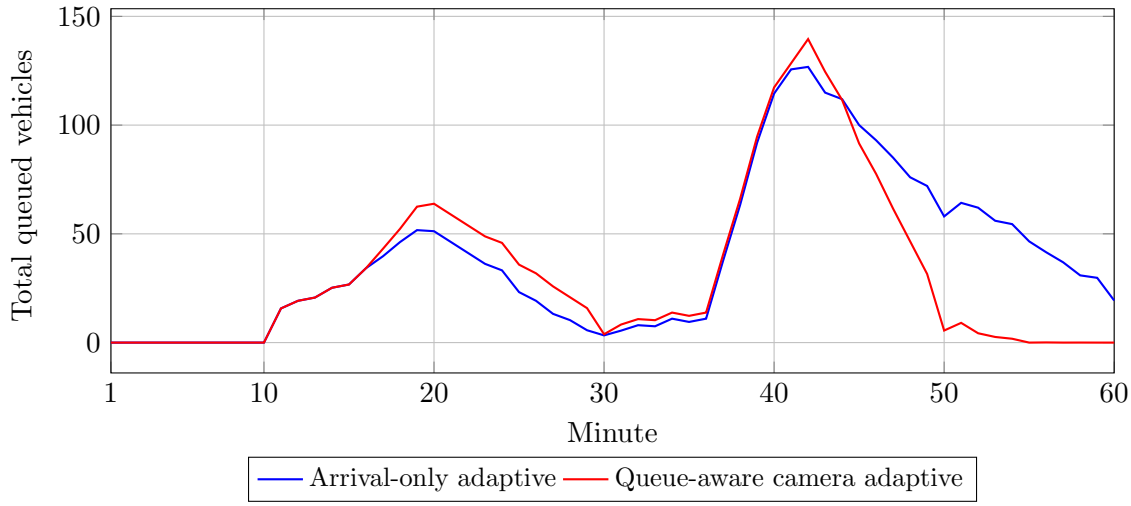


Figure 4: Total queue under the two adaptive controllers. Queue awareness becomes most useful after abrupt demand changes and temporary losses in discharge.

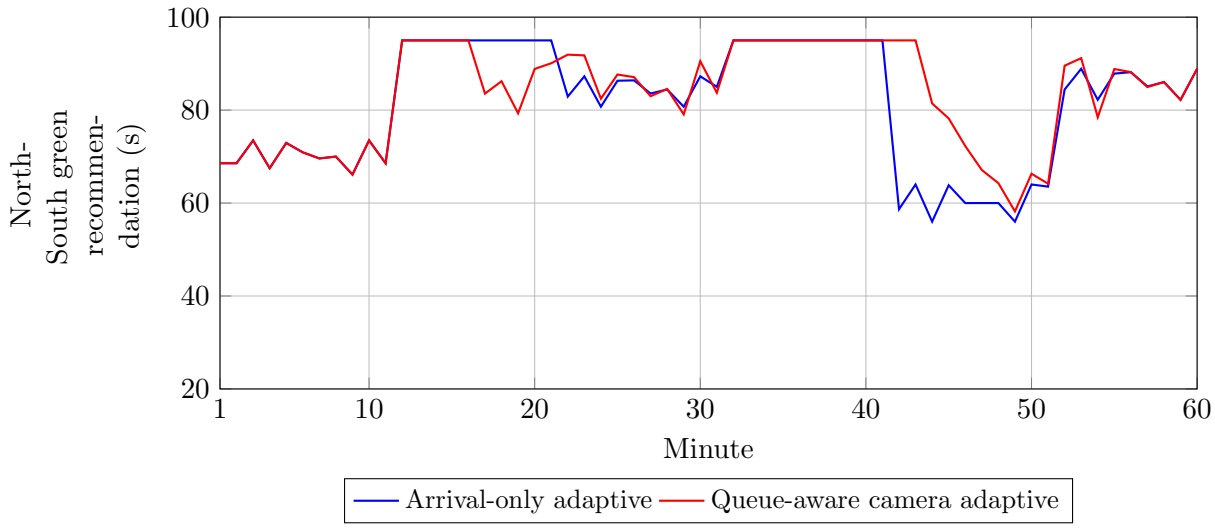


Figure 5: Recommended North-South green time. The two controllers diverge when inherited queues and measured discharge no longer match the newest arrival ratio.

The advantage of the queue-aware method comes from the information used by the controller. Queue is a state variable that persists, service is explicitly modeled, and short-term changes create consequences in later minutes. The controller therefore responds not only to the newest arrivals but also to congestion that has already accumulated and to how effectively each direction is clearing.

7. Interpretation and Evanston Relevance

The strongest direct conclusions are the modeled delay reductions produced when the observed traffic measurements are passed through the signal-control models. Converting those reductions into fuel, emissions, or monetary values requires additional assumptions, so those estimates remain secondary.

For the Phase I morning observation, the fixed optimization saves approximately 8,782 vehicle-seconds, or 2.44 vehicle-hours, in the model. Using the original project's occupancy and value-of-time assumptions gives roughly 3.61 person-hours of saved time and about $76 in transportation time value for that hour. Fuel and carbon effects can also be estimated from avoided idling, but they are derived estimates rather than quantities directly measured at the intersection.

The local context is particularly relevant. Evanston's Green Bay Road Corridor Improvements Project includes traffic-signal improvements at Lincoln Street, the same intersection examined here, and the City's project documentation identifies signal modernization as part of the corridor work [2, 3]. The City's 2026 capital-project summary also describes Green Bay Road as an important north-south route used by pedestrians, buses, and vehicles and notes ongoing signal modernization along the corridor [4].

This does not mean the model should be applied directly to the operating signal. Municipal timing must account for pedestrian clearance, turning movements, coordination with neighboring signals, safety standards, transit, and site-specific saturation flow. The value of the project is narrower and more defensible: it shows, using a real Evanston field observation, how much information is lost when a traffic signal is treated as a fixed equal split and how queue-aware sensing can change a timing recommendation when the current traffic state moves away from the hourly average.

8. Limitations

The study uses real field observations, but the controller and delay calculations remain simplified models of an operating traffic signal.

- **Single observation period.** The main results come from one one-hour morning observation at one intersection. Repeating the study across multiple weekdays, seasons, and weather conditions would show how stable the findings are.
- **Simplified service capacity.** The baseline value $\mu = 1.0$ vehicle/second is a combined phase-pair discharge assumption rather than a full lane-by-lane saturation-flow calibration.
- **Two-phase geometry.** Protected turns, pedestrian phases, bicycles, buses, emergency preemption, and lane-by-lane interactions are not explicitly represented in the headline controller.
- **No corridor coordination.** Green Bay Road contains multiple signalized intersections. The model optimizes Lincoln Street and Green Bay Road independently and does not yet optimize signal offsets with neighboring intersections.
- **Camera performance is not experimentally evaluated here.** The field variables were observed directly. A deployed vision system would introduce additional issues such as occlusion, nighttime visibility, weather, camera angle, and detection error.
- **The 400-ft observation distance is a design baseline.** A real installation would set the detection zone from actual intersection geometry, sight lines, and camera placement.

These limitations are important because the project evaluates a control concept, not a complete municipal signal design. The results are best interpreted as evidence that observed directional imbalance, queue buildup, and discharge conditions can materially change the timing recommended by the simplified model.

9. Future Work

The most useful next step is replication. Additional morning and evening observations at Lincoln Street and Green Bay Road would show whether the measured directional imbalance and short-term queue patterns are consistent across days. The same method could then be applied to other intersections along the Green Bay Road corridor.

A second step is to implement the camera pipeline itself. Recorded or live video could be processed to count vehicles entering the observation zone, estimate queues, and measure vehicles crossing the stop line during green. Those automatically extracted measurements could be compared against the manually recorded field data to quantify detection error.

A third step is to compare the model with professional signal operation. Evanston's Green Bay Road project specifically includes signal improvements at Lincoln Street [2]. If existing morning timing plans or engineering assumptions are available from the City, the model could be run in shadow mode alongside those plans without changing the operating signal. A more advanced version would also include turning movements, pedestrian constraints, lane-level saturation flow, and offsets between neighboring signals.

10. Conclusion

This project began with a simple local question: if substantially more vehicles are traveling through an intersection in one direction, should both directions receive equal green time? A one-hour morning observation at Lincoln Street and Green Bay Road recorded 3,122 vehicle arrivals, with 71.65% traveling North-South. Under the simplified Phase I model, that distribution shifts a 120-second cycle from 60/60 to approximately 85.98/34.02 and reduces expected red-signal delay by 18.75%.

Phase II addresses what the hourly average cannot show. The minute-level field record contains changing arrival rates, accumulated queues, and variations in discharge. By carrying queue state from one minute to the next, the adaptive model can react to congestion that an arrival-only controller misses. Under the Phase II queue-delay framework, the queue-aware method produces 16.77% less cumulative queue delay than the arrival-only adaptive controller and 42.02% less than the optimized fixed plan.

The project therefore supports a two-part conclusion. Long-run directional demand can be used to improve a poorly matched fixed split, while short-term queue and discharge information can improve the response when current conditions move away from that average. This is directly relevant to Evanston because Lincoln Street is included in the City's Green Bay Road traffic-signal improvement work. The next step is not to treat the model as a replacement for municipal engineering, but to compare it with additional field observations and the constraints used in Evanston's actual signal-design process.

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A. Computational Summary

Phase II is evaluated for sixty one-minute intervals. At each interval:

1. read the newest North-South and East-West arrival counts;
2. carry forward the current queues from the previous interval;
3. determine the signal split for each strategy;
4. calculate vehicles served from green time and measured discharge effectiveness;
5. update the queues using

$$Q_d(t+1) = \max(0, Q_d(t) + A_d(t) - S_d(t));$$

6. accumulate queue delay using the average queue during the 60-second interval.

The observed minute-level arrival sequence sums to the same one-hour totals used in Phase I: 2,237 North-South vehicles and 885 East-West vehicles. Both adaptive controllers use the preceding interval's measurements for the next timing recommendation, so neither controller uses future information.

B. Phase II Minute-Level Field Observations

Min.	NS	EW	Min.	NS	EW
1	28	21	31	48	9
2	30	19	32	50	7
3	27	21	33	47	10
4	31	20	34	51	8

Min.	NS	EW	Min.	NS	EW
5	26	18	35	46	6
6	29	21	36	49	9
7	28	20	37	48	8
8	27	22	38	47	10
9	30	19	39	50	7
10	24	18	40	44	6
11	50	9	41	22	23
12	51	7	42	24	21
13	49	10	43	21	24
14	52	8	44	25	22
15	48	6	45	20	20
16	51	9	46	23	23
17	50	8	47	22	22
18	49	10	48	21	24
19	51	7	49	24	21
20	46	6	50	18	16
21	38	17	51	38	16
22	40	15	52	40	14
23	37	18	53	37	17
24	41	16	54	41	15
25	36	14	55	36	13
26	39	17	56	39	16
27	38	16	57	38	15
28	37	18	58	37	17
29	40	15	59	40	14
30	34	14	60	34	13

How Road Geometry Changes Adaptive Signal Performance

Nine hypothetical intersection types, evaluated alongside the original Evanston field study

The preceding 16-page paper is preserved unchanged. This appendix extends the research question from one observed intersection to a controlled set of **hypothetical** configurations. None of the new traffic counts or performance outcomes are presented as Evanston field measurements.

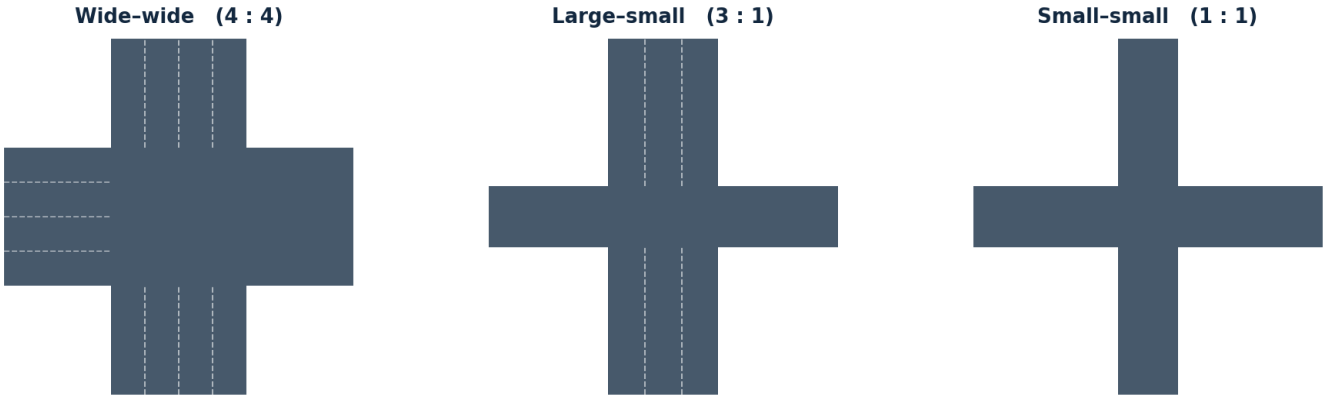


Figure A1. Illustrative plan views for wide–wide, large–small, and small–small crossings. Widths are schematic, not scale drawings.

1 Research question

Does access to current queues remain useful when the two approaches have different widths and lane capacities? A controller that works at one predominantly north–south intersection may behave differently when each road has similar demand or when a major road intersects a low-volume side street.

2 Study design and assumptions

The original paper uses one hour of recorded arrivals at Lincoln Street and Green Bay Road, followed by modeled comparisons. Here, nine **synthetic** road configurations are tested under a deliberately simplified, repeatable 60-minute demand sequence. The categories refer to the number of effective lanes serving a phase pair, not the entire cross-section of a real road.

All cases use two conflicting phase groups, a 120-second allocatable cycle, green limits of 25–95 seconds per phase, and a nominal service capacity of **0.45 vehicle per second per effective lane** during green. The value is an illustrative assumption, not a locally measured saturation rate. The study does not include clearance intervals, pedestrian minimums, turns, transit priority, spillback, or network coordination.

Each synthetic arrival stream contains time-varying waves and two temporary directional surges. All four controllers receive the **same** arrivals and service assumptions. One-minute decisions use the preceding minute’s information. The queue-aware variant uses the current backlog plus recent arrivals, adjusted for recent observed discharge, to apportion green time.

3 Tested intersection categories

Wide–wide (4 vs. 4 effective lanes): major / major; nominal arrivals 36.6 vs. 36.6 vehicles/minute.

Large–large (3 vs. 3 effective lanes): arterial / arterial; nominal arrivals 30.0 vs. 28.8 vehicles/minute.

Large–medium (3 vs. 2 effective lanes): arterial / collector; nominal arrivals 33.6 vs. 21.6 vehicles/minute.

Large–small (3 vs. 1 effective lanes): arterial / local; nominal arrivals 34.8 vs. 12.6 vehicles/minute.

Medium–medium (2 vs. 2 effective lanes): collector / collector; nominal arrivals 21.6 vs. 20.4 vehicles/minute.

Medium–small (2 vs. 1 effective lanes): collector / local; nominal arrivals 25.8 vs. 12.0 vehicles/minute.

Small–small (1 vs. 1 effective lanes): local / local; nominal arrivals 12.0 vs. 10.8 vehicles/minute.

Wide–narrow (4 vs. 1 effective lanes): wide / narrow; nominal arrivals 43.2 vs. 11.4 vehicles/minute.

Narrow–narrow (1 vs. 1 effective lanes): single-lane / single-lane; nominal arrivals 12.0 vs. 9.6 vehicles/minute.

4 Modeled comparisons

The primary table compares the fixed split based on **average directional demand** with the adaptive rule using real-time queues. Mean delay is cumulative modeled vehicle-seconds divided by total synthetic arrivals. Negative reductions would indicate that the adaptive controller performed worse.

Road configuration	Lanes A/B	Fixed delay (s/veh)	Queue-aware (s/veh)	Change
Wide–wide	4 / 4	14.3	4.8	+66.7%
Large–large	3 / 3	26.0	8.8	+66.3%
Large–medium	3 / 2	95.4	32.6	+65.8%
Large–small	3 / 1	239.7	74.2	+69.0%
Medium–medium	2 / 2	44.2	18.9	+57.1%
Medium–small	2 / 1	212.8	97.3	+54.3%
Small–small	1 / 1	72.1	44.4	+38.5%
Wide–narrow	4 / 1	215.1	32.7	+84.8%
Narrow–narrow	1 / 1	50.7	26.0	+48.7%

Table A1. Scenario-only outputs. These should not be mixed with the original 42.02% reduction, which comes from the separate measured-data study.

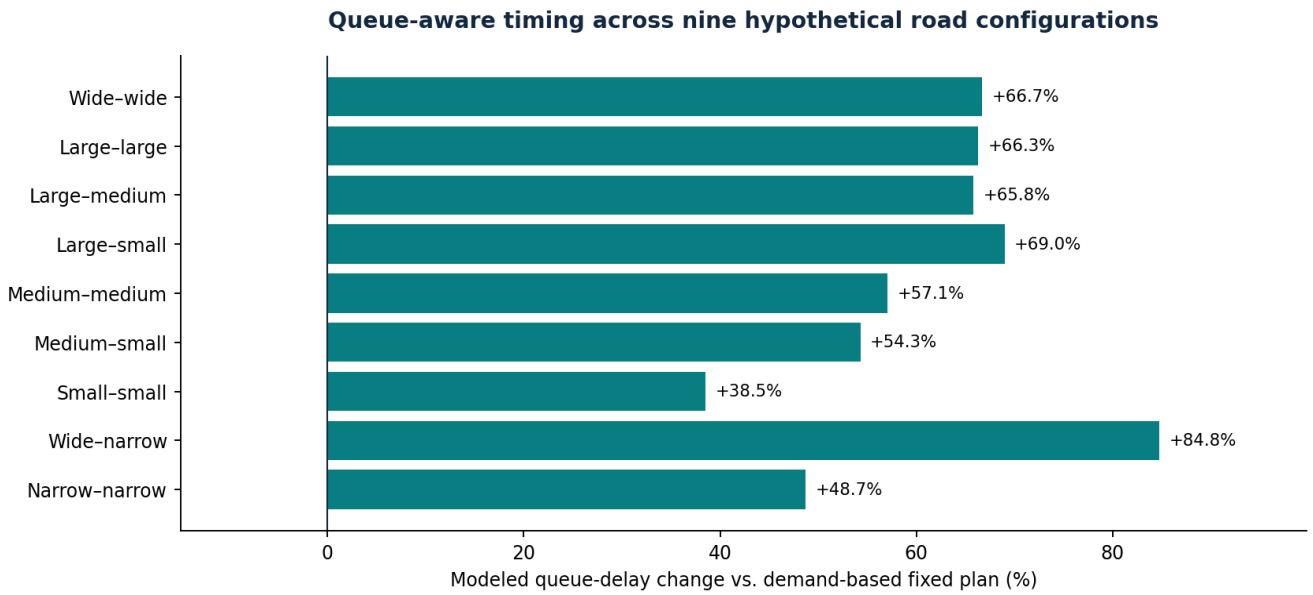


Figure A2. Percent change in modeled delay versus the demand-optimized fixed baseline. Results depend on the synthetic arrival waves and capacity assumptions.

5 Within-hour queue behavior

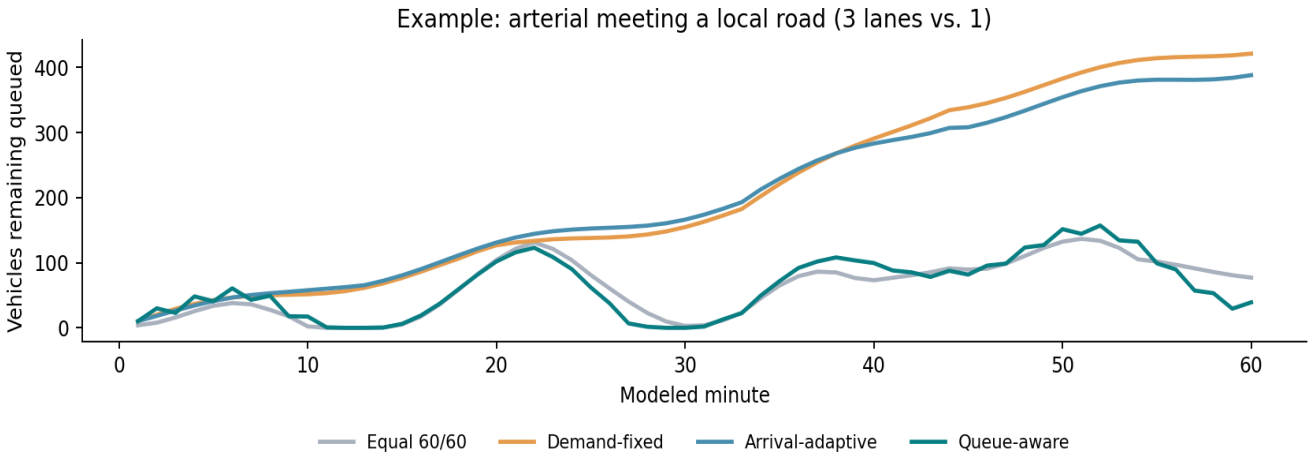


Figure A3. Four timing strategies applied to the same hypothetical large–small intersection. Unlike a single headline percentage, the trajectories show when backlogs build and clear.

6 What changes with geometry?

Major–major crossings. When both approaches offer multiple effective lanes, greater green-time allocation can serve larger incoming flows. Queue-aware control is most relevant when a short-lived imbalance emerges; extra green does not necessarily help if the opposing movement has comparable demand.

Major–minor crossings. A low-volume side street may not justify the same green share as a busy arterial. However, a queue on the small road can still become important, particularly when its discharge capacity is limited. Simple arrival-only allocation can miss that inherited backlog.

Minor–minor crossings. Small absolute arrivals can produce little delay in all strategies. A large percentage improvement might represent a modest number of seconds, so reporting both relative and absolute effects is essential.

Wide–narrow crossings. Lane count and demand must be treated separately. Road width alone is not enough to choose the split: a wide but quiet approach may deserve less green than a narrow but temporarily congested one.

7 Sensitivity and practical deployment

These synthetic experiments are **comparative demonstrations**, not validated traffic-engineering forecasts. Their results may change substantially if one modifies arrival variability, lane-level saturation flows, turning movements, pedestrian green minima, detector noise, cycle length, or driver behavior. The original study's simplifying assumptions still apply.

A realistic next step would be to obtain lane geometry, turning-movement counts, signal phasing, pedestrian timings, and existing plans for several Evanston intersections; calibrate each case independently; then test the controller in shadow mode before any field deployment. A city engineer could help identify representative major–major, major–minor, and minor–minor study sites.

8 Connection back to the original study

The original field-observation analysis at Lincoln Street and Green Bay Road recorded 3,122 arrivals. It found **18.75%** lower expected red-signal delay for an optimized fixed split than for a 60/60 split (Phase I). Under the separate persistent-queue framework, the camera-aware controller produced **42.02%** lower modeled cumulative queue delay than the optimized fixed plan and **16.77%** less than the arrival-only controller (Phase II). Those figures remain unchanged. The nine synthetic studies broaden the hypothesis without claiming nine additional field validations.

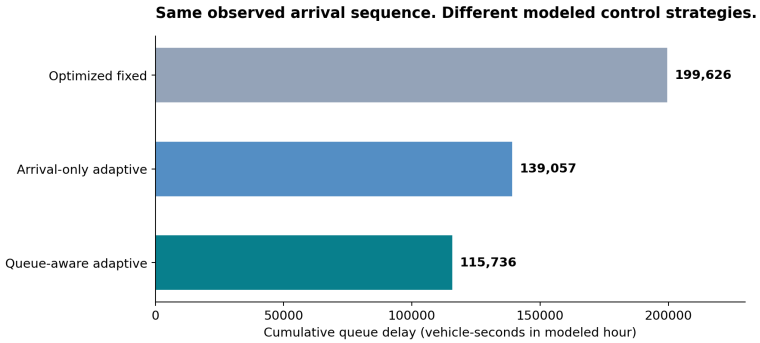
Methods note. Reproducible definitions: 60 intervals of 60 seconds; initial queues zero; effective green share clipped to 25–95 of 120 seconds; served vehicles = min(available vehicles, 60 × 0.45 × effective lane count × green share); next queue = max(0, previous queue + arrivals − served); delay = sum over intervals of 30 × (starting queue + ending queue), both directions. Synthetic arrival multipliers use deterministic sine waves and specified temporary surges. No randomness, field camera detections, or additional measured intersection datasets are used.

What Could Reduced Queuing Be Worth?

An economic sensitivity and scaling supplement to the 21-page research report. Calculations are illustrative and not predictions of real savings. This appendix preserves the source study and its previous nine hypothetical intersection scenarios.

Measured input vs modeled output

The field-observation input consists of 3,122 arrivals in one morning hour at Lincoln Street and Green Bay Road. The research report models 199,626 vehicle-seconds of queue delay with the optimized fixed plan versus 115,736 vehicle-seconds with queue-aware timing. The modeled difference is **83,890 vehicle-seconds = 23.30 vehicle-hours in that hour**, or 42.02% relative to the optimized fixed comparator. These are simulated delays, not observed travel-time savings from a deployed traffic controller.



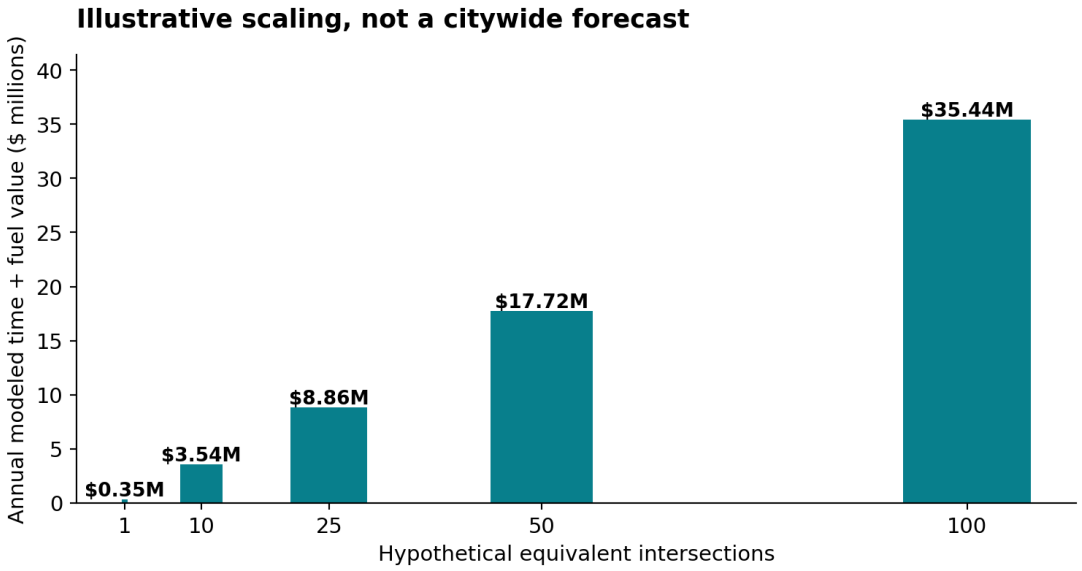
Converting modeled time into illustrative economic value

Assumptions: 1.4 persons per vehicle (not observed); $21.10 per person-hour (USDOT 2025 BCA guidance, all-purpose travel value in 2023 dollars); 0.25 gasoline gallons per vehicle-hour of reduced idling (illustrative, not measured); $3.50 per gallon (illustrative); 8.887 kg CO■ per gallon combusted (US EPA). This calculation treats reduced modeled queue-delay hours as potentially avoided idle-hours, a simplifying proxy that may substantially overstate fuel or emissions benefits.

Per modeled hour	Calculation	Illustrative estimate
Vehicle delay avoided	(199,626 - 115,736) / 3,600	23.30 vehicle-hours
Person time avoided	23.30 × 1.4	32.62 person-hours
Value of time	32.62 × $21.10	$688.36
Fuel proxy	23.30 × 0.25	5.83 gallons
Fuel cost proxy	5.83 × $3.50	$20.39
Total illustrative value	Time value + fuel cost	$708.75
CO■ proxy	5.83 × 8.887	51.77 kg

From One Intersection to a Network

The visual below scales the **same modeled hourly difference** to multiple identical-equivalent intersections, each operating 2 comparable peak hours/day on 250 days/year (=500 hours/year). This is an intentionally simple what-if exercise. Other intersections will not share the same traffic or performance, and real benefits cannot be inferred by multiplication.



Illustrative annual scenarios

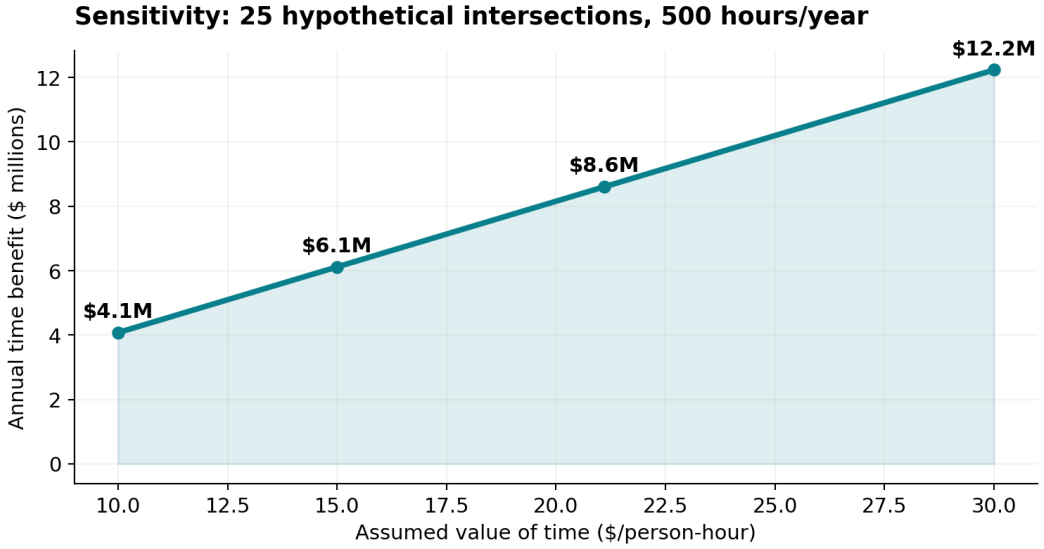
Equivalent sites	Time + fuel value	CO■ proxy
1	$0.35 million	25.9 metric tons
10	$3.54 million	258.9 metric tons
25	$8.86 million	647.2 metric tons
50	$17.72 million	1,294.3 metric tons
100	$35.44 million	2,588.6 metric tons

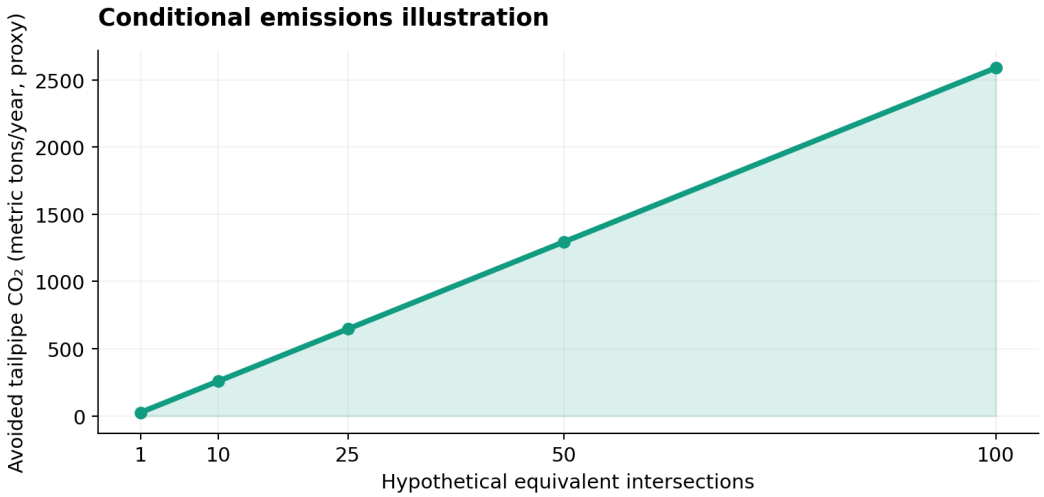
How to interpret large numbers

Not an annual forecast. These represent a hypothetical network of matched intersections with identical performance, assumed travel valuations, and sustained peak-hour benefits. They omit capital/maintenance costs, safety impacts, differing congestion levels, demand feedback, equipment reliability, pedestrian phases, and corridor coordination. They should never be labeled “Evanston savings” or “city money saved.”

How Sensitive Are the Estimates?

Sensitivity is essential because monetary and environmental figures are not field observations. The economic estimate increases linearly with the assumed number of people per vehicle, the assigned value of time, and the number of comparable operating hours.





Conversion equations: Δvehicle-hours = (199,626 - 115,736)/3,600; annual time benefit = Δvehicle-hours × people/vehicle × $/person-hour × comparable hours/year × sites; annual fuel proxy = Δvehicle-hours × gallons/idle-hour × hours/year × sites; annual CO■ proxy = fuel gallons × 8.887 kg/gallon. These proxy calculations do not simulate engine operation, acceleration, vehicle fleet composition, or actual emissions.

Evidence and external factors

Research report: pages 10–13 (Phase II queue-delay model and limits), pages 16–21 (nine explicitly hypothetical intersection geometries). External: US DOT, Benefit-Cost Analysis Guidance 2025 Update, Table A-2 ($21.10 per person-hour, 2023 dollars), https://www.transportation.gov/sites/dot.gov/files/2024-11/Benefit%20Cost%20Analysis%20Guidance%202025%20Update%20%28Final%29.pdf ; US EPA Greenhouse Gas Equivalencies Calculator: https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator-calculations-and-references (8.887 kg CO■/gallon). Occupancy, idle rate, fuel price, and deployment scale are explicitly illustrative choices, not sourced field measurements.